PROBABILITY AND STATISTICS
PROJECT REPORT

Predicting Internet Advertisements

From Image Geometry and Web-Context Features

A reproducible study of the UCI Internet Advertisements dataset

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1 Data overview and research objectives

1.1 Problem, scope, and principal findings

The task is to predict whether a candidate image on a web page is an advertisement (`ad`) or not (`nonad`). The supplied file is not an image collection: it is a tabular representation containing image geometry and indicators extracted from surrounding web context. Consequently, this study evaluates classification from supplied features, not visual recognition from pixels.

The selected full-feature ridge logistic model achieved held-out average precision (AP) of 0.863 and ROC AUC of 0.957. At a threshold selected exclusively on validation data, it detected 67 of 84 test advertisements, with 25 false alarms among 542 non-advertisements. Test F1 was 0.761. The main finding is that web-context features substantially improve ranking beyond geometry alone, while the best model for ranking is not necessarily the best classifier at a particular threshold.

1.2 Provenance and unit of observation

The local `data/add.csv` contains 3,279 rows, 1,558 predictors, one class column, and an extra exported row index. Every predictor and label was compared with the original UCI `ad.data` download: numeric values, missingness positions, and labels match in row order. Feature names were recovered in order from `ad.names`. The source is credited to Nicholas Kushmerick; the repository identifies the dataset as Internet Advertisements, DOI 10.24432/C5V011 [2].

The original study is Kushmerick (1999), *Learning to remove Internet advertisements* [3], Proceedings of the Third Annual Conference on Autonomous Agents, pp. 175–181, DOI `https://doi.org/10.1145/301136.301186`. The source documentation identifies the supplied matrix as its STANDARD encoding and reports historical C4.5rules accuracy above 97%. That result is not a replication target here: the original evaluation protocol has not been recovered reliably enough for a like-for-like comparison.

One row represents a candidate image described through extracted attributes. The accompanying documentation dates data generation to April–July 1998. The supplied files contain measurements, encoded context indicators and class labels rather than the original images and pages. The available collection details and their implications are discussed in Section 6.

The observed class counts are 459 ads (14.00%) and 2,820 nonads (86.00%). The accompanying original documentation instead states 458 and 2,821. The actual local and original data agree with each other, so all calculations use the measured counts, not the inconsistent prose. This discrepancy is documented rather than silently corrected in the source.

1.3 Research questions

1. How do the available geometric measurements differ between ad and nonad observations, and what does missingness imply for their interpretation?
2. Do the full context features improve held-out ranking beyond geometry and locality alone?
3. How do ranking, probability quality, and classification trade-offs differ among a majority baseline, ridge logistic models, and a constrained decision tree?
4. How uncertain are the final test estimates when repeated predictor patterns are treated as clusters, and what limits generalization?

The analysis answers these questions using withheld predictor-pattern groups: records with identical predictors stay together throughout fitting and evaluation.

Table 1: Predictor families recovered from the original ordered dictionary.

Family	Count	Meaning / representation
Height, width, aspect ratio	3	Continuous geometric measurements
Local	1	Binary locality indicator; exact operational rule unavailable
URL terms	457	Binary phrase indicators from image URL
Original-URL terms	495	Binary indicators in source `origurl` family
Anchor-URL terms	472	Binary phrase indicators from anchor URL
Alt-text terms	111	Binary phrase indicators from alternative text
Caption terms	19	Binary phrase indicators from nearby/caption context
Total	1,558	Three continuous and 1,555 binary predictors

Table 1 shows that contextual indicators dominate dimensionality. For example, a row combines height, width and aspect ratio with indicators such as `url*ads`. A value of one records the presence of the corresponding encoded URL feature and zero its absence; it is not a word count. The source dictionary also includes multi-term names such as `url*images+buttons`. Thus each row describes both an image's shape and features of its surrounding web context. A phrase indicator denotes the source encoding, not a directly verified linguistic or causal mechanism. The dictionary names geometry but does not provide explicit units; numerical dimensions are therefore reported in source units, rather than assigning undocumented units.

2 Statistical and predictive background

2.1 Binary outcomes, loss, and regularized logistic regression

Let $Y_i = 1$ denote ad and X_i the observed feature vector. Logistic regression models

$$p_i = P(Y_i = 1 \mid X_i) = \{1 + \exp[-(\beta_0 + X_i^T \beta)]\}^{-1}.$$

For standardized fitting inputs Z, write $p_i = \text{logit}^{-1}(\gamma_0 + Z_i^T \gamma)$. Ridge regression estimates γ by minimizing the mean binomial negative log-likelihood plus a quadratic penalty,

$$-\frac{1}{n} \sum_i \{y_i \log p_i + (1 - y_i) \log(1 - p_i)\} + \frac{\lambda}{2} \sum_j \gamma_j^2.$$

Here γ denotes fitting-scale slopes and β denotes the back-transformed slopes reported later. The intercept is not penalized. In `glmnet`, ridge corresponds to $\alpha = 0$ and standardization is performed using the fitting sample [1]. Regularization is important because the predictor count is large relative to the training sample and binary indicators can be rare or correlated. A finite ridge solution is more suitable than attempting unrestricted maximum-likelihood inference across all 1,558 predictors.

This is a predictive model: coefficients describe conditional fitted associations, not causal effects. Standard unpenalized coefficient p-values are not supplied for this penalized, tuned model. Non-linearity and interactions may be inadequately represented by a linear logit; regularization does not prove model correctness or probability calibration.

The geometry model uses $\log(1 + x)$ transformations of height, width, and aspect ratio, the locality indicator, and four missingness indicators. The full model adds all context indicators. This comparator is technically geometry-plus-locality; the shorter label "geometry" refers to that explicit feature set.

2.1.1 Why this model and scale?

The model is linear in the *log odds*, $\log\{p/(1-p)\}$, not in probability. A one-unit increase in a transformed predictor changes fitted odds by $\exp(\beta_j)$, holding the other encoded inputs fixed. A positive score contribution raises probability most near the middle of the logistic curve and less near its extremes. This distinction explains why coefficient signs alone do not imply a useful classification threshold.

The log1p transform compresses large geometric values while retaining zero. A change from source width w_1 to w_2 contributes $\beta_w \log\{(1+w_2)/(1+w_1)\}$ to the geometry-model logit. It therefore makes proportional differences more comparable across scales, without assuming that geometry is normally distributed. It is a modeling choice, not a normality correction; logistic regression does not require normally distributed predictors.

With $z_{ij} = (x_{ij} - \mu_j)/s_j$, the fitting-scale ridge penalty shrinks standardized slopes γ_j. Back-transformation gives $\beta_j = \gamma_j/s_j$ and adjusts the intercept. Thus its penalty on the reported-scale slopes is proportional to $\sum_j s_j^2\beta_j^2$, not an equal penalty on every raw-scale coefficient. For a binary feature of frequency q, $s_j^2 = q(1-q)$: a large raw slope for a rare indicator need not mean a large standardized contribution. Strong shrinkage approaches a near-constant probability; weak shrinkage permits a more variable fitted surface. The selected penalty trades training fit against complexity through development data, rather than proving that the linear-logit representation is true.

2.2 Constrained classification tree

A classification tree partitions feature space through sequential splits and assigns probabilities using terminal-node class frequencies. For binary classes, the default Gini impurity at a node with ad fraction q is $G(q) = 2q(1-q)$. A candidate split gains $G(q) - \{n_L G(q_L) + n_R G(q_R)\}/n$. This is a local improvement, not the AP criterion used to compare the completed trees. Sequential splits can express interactions: a URL indicator can matter in one width region and not another. Depth and minimum-node constraints limit that flexibility but do not establish stability. The `rpart` implementation uses classification splitting and a complexity parameter *cp*; larger *cp* discourages splits with small improvements [4]. The tree here has maximum depth five and minimum split size 20. It provides a nonlinear comparator without building an extensive model-search exercise. Leaf probabilities are discrete and can be poorly calibrated; a strong classification threshold need not imply strong ranking across the entire score range.

2.3 Metrics and decision thresholds

An advertisement is the positive class. A true positive (TP) is an ad correctly classified as ad; a false positive (FP) is a nonad incorrectly classified as ad. A false negative (FN) is a missed ad, and a true negative (TN) is a correctly classified nonad. Precision measures the fraction of predicted ads that are actually ads; recall measures the fraction of actual ads detected. Their harmonic mean, F1, balances these two fractions:

$$\text{precision} = \frac{TP}{TP+FP}, \quad \text{recall} = \frac{TP}{TP+FN}, \quad F_1 = \frac{2TP}{2TP+FP+FN}.$$

Balanced accuracy averages sensitivity and specificity. Accuracy alone is insufficient: predicting every case as nonad already scores about 86% here while detecting no ads. When no positive predictions occur, precision is recorded as zero by reporting convention, not as an estimated conditional probability.

ROC AUC is the probability that a random positive receives a higher score than a random negative, counting ties as one half. AP aggregates precision over recall increments at distinct score thresholds:

$$\mathrm{AP} = \sum_k (R_k - R_{k-1}) P_k.$$

Ties are pooled before computing increments. AP is not trapezoidal PR area [5], and depends on class prevalence. A constant-score classifier has AP equal to observed positive prevalence. The Brier score $n^{-1} \sum_i (p_i - y_i)^2$ evaluates squared probability error; smaller is better. Calibration means agreement between predicted probabilities and observed frequencies: among many candidates assigned probability .8, roughly 80% should be ads. A reliability plot compares these quantities within score bins. Brier score assesses overall probability error, so it reflects both calibration and the ability to distinguish classes.

Model selection maximizes training cross-validation AP. The final decision threshold maximizes validation F1, choosing the highest threshold if tied. AP selects ranking ability, whereas F1 selects one operating point; their preferences need not coincide. F1 gives precision and recall equal emphasis; Section 6 examines how the observed trade-off changes when false positives and missed ads have different costs.

2.4 Metric meaning and prevalence

A ranking asks which candidate to inspect first; a probability asks how often similarly scored candidates are ads; a decision asks whether the consequences justify acting. AP evaluates the first, Brier assesses squared probability error, and F1 summarizes one decision set. For a four-record illustration with descending labels $(1, 0, 1, 0)$, positive records arrive at recalls $1/2$ and 1 with precisions 1 and $2/3$, giving AP $\frac{1}{2}(1) + \frac{1}{2}(2/3) = 5/6$. Moving probabilities without changing their order leaves AP unchanged but can change Brier and thresholded decisions.

If sensitivity is s, false-positive rate is f, and ad prevalence is π, precision equals

$$\frac{\pi s}{\pi s + (1 - \pi) f}.$$

The frozen test rates $s = 67/84$ and $f = 25/542$ give observed precision .728 at prevalence $84/626$, but about .149 at hypothetical prevalence .01. This calculation holds sensitivity and false-positive rate fixed and illustrates why precision changes with the class mix.

2.5 Exploratory distribution comparisons and uncertainty

The rank comparison $U/(n_1 n_0)$ summarizes the fraction of observed ad–nonad pairs for which the ad has a larger value, counting ties as one half. This is a descriptive, row-weighted measure of separation, not automatically a median effect. Missing measurements are excluded separately for each variable. Because repeated representations and unobserved website dependence undermine an independent-row reference distribution, geometry significance tests are not used as evidence in this report. The analysis exports descriptive rank separation without independent-row significance tests.

Predictive inference instead uses an explicit working model: representation clusters are exchangeable draws from a benchmark-like generating mechanism, and each cluster contributes all of its rows. The target weights candidate-image records, rather than unique signatures equally. More precisely, the inferential target is the AP difference between these two fixed scoring functions under the row distribution induced by independent draws of representation clusters from the working population. It is not the difference between algorithms retrained on new datasets. The

observed finite-test difference is a separate descriptive quantity; the interval concerns sampling that test composition under this model. Under that model a population metric is a functional of cluster-level counts and scores; for example recall is $E[TP_g]/E[TP_g + FN_g]$, where g denotes a cluster. The ratio is estimated by totals over the sampled clusters. This model permits arbitrary dependence within a signature but assumes independence across signatures, an assumption not established for shared websites. It is a conditional modeling abstraction, not a claim about a probability sample of Internet images. The bootstrap allows cluster composition and prevalence to vary; it is not design-based inference conditional on the fixed stratified allocation.

For final predictive uncertainty, 1,000 bootstrap samples resample entire test predictor-pattern groups with replacement. The number sampled equals the 484 observed test groups; the row count can vary because group sizes differ. Each replicate recomputes the whole metric, rather than averaging per-group F1 or AP. For the AP contrast, $\Delta_b = \mathrm{AP}_{full,b} - \mathrm{AP}_{geometry,b}$ uses the same sampled groups in both terms, preserving their shared easy and difficult cases. All rows in a selected group are retained, and a repeatedly selected group contributes repeatedly. Percentile endpoints at 2.5% and 97.5% quantify conditional test-sample uncertainty. Training, parameter selection, and threshold selection are not repeated; intervals do not include their instability or unobserved domain clustering.

3 Data preprocessing and evaluation design

3.1 Observed integrity checks

The exported index runs consecutively from 0 to 3,278 and is removed before modeling. It is not a predictor. Class strings are trimmed, terminal punctuation is normalized conceptually, and ad is encoded as one. The original ordering places many ads together; neither an order-based split nor using the index would be appropriate.

Numeric parsing recognizes whitespace-padded question marks as unknown values. Missingness occurs in height (903 rows), width (901), aspect ratio (910), and locality (15). At least one geometry value is absent in 910 rows; 920 rows have any missing predictor. Thus 28.06% of rows have missing data. The source's "unknown" interpretation is retained; unknown is not converted to zero.

All observed non-geometric predictors are binary. Their mean observed activation rate is 0.821%, illustrating sparsity. No globally constant source feature was found. Valid extreme dimensions are retained: an outlier rule is not evidence of an incorrect image size. The original file remains unchanged.

3.2 Duplicate-pattern audit

There are 2,420 distinct predictor patterns and 859 repeated rows beyond their first occurrences (26.20% of rows). Pattern identity includes all predictors and missingness positions but excludes labels and the exported index. These are identical *representations*; the data do not establish whether they are literally the same image or page.

Three patterns have conflicting labels: group sizes and ad counts are respectively $(49, 1)$, $(3, 2)$, and $(2, 1)$. The smallest unavoidable number of errors for any deterministic classifier that sees only this feature representation is three on these observed rows. This is an in-sample representation constraint, not an estimate of a population Bayes error. Conflicting labels are not changed or deleted. The first two conflicting groups fall in training and the third in validation; none is in the test partition.

A random row split could place indistinguishable representations on both sides of evaluation. To avoid that, all rows with the same predictor pattern are assigned to the same partition and training fold. Different patterns can nevertheless share pages or domains: unavailable identifiers prevent stronger leakage controls.

Table 2: Fixed grouped partition, with class counts measured after allocation.

Partition	Rows	Patterns	Ads	Nonads
Train	1984	1452	288	1696
Validation	669	484	87	582
Test	626	484	84	542

Table 2 reflects approximately 60/20/20% of patterns, not exactly those percentages of rows. Groups are stratified by majority label (ties assigned to ad); within each stratum their allocation is randomized. The delivered split assignments identify the evaluation sample; robustness to alternative allocations is not estimated. Labels are used for stratification, not to compute feature transformations. The partition is a designed benchmark comparison, not a guarantee of naturally occurring deployment prevalence.

3.3 Training-only transformations and model selection

Training, validation and test data have distinct roles. Training data estimate coefficients, tree splits and preprocessing values. Five-fold cross-validation repeatedly fits on four training folds and scores the fifth; the held-out AP values guide parameter and model selection. Validation data choose the decision threshold for the selected fitted model. Test data then estimate its performance on observations not used for either fitting or selection.

The three geometry variables undergo $\log(1+x)$ transformation. Four missingness indicators are appended. Medians are learned separately from each training fold, and used to fill corresponding fold-held-out data. For binary locality, the median is a modal value here. No class balancing, oversampling, or synthetic rows are introduced.

The five-fold training procedure keeps identical patterns together. Ridge models examine 25 logarithmically spaced penalties from 10^2 to 10^{-4}; the tree examines $cp \in \{.001, .005, .01, .02, .05\}$. Each model's parameter maximizes the unweighted mean of five fold AP values. The winning model family is also selected by this training-only criterion. Mean fold AP need not equal AP of pooled out-of-fold predictions.

After selection, each comparator is fit on the full training partition. Medians and ridge scaling remain training-derived. The selected model's threshold is selected on validation data only, then frozen before test metrics are generated. Training and validation are *not* merged for a new final fit; this preserves the exact fitted model associated with the selected threshold. The model-selection CV scores are development scores, not unbiased final-performance estimates.

3.4 Reproducibility

The delivered split assignments, fold IDs, fitted models, predictions, and analysis scripts identify the actual experiment. Diagnostic analyses added after inspecting test results are labeled as post-evaluation descriptions and do not change the fitted model or threshold.

4 Descriptive statistics

Table 3: Observed geometry by class: median [first quartile, third quartile], excluding unknown values.

Variable	Class	Observed n	Missing n	Median [Q1,Q3]
Height	nonad	1990	830	43.000 [23.000, 90.000]
Height	ad	386	73	60.000 [60.000, 60.000]
Width	nonad	1992	828	101.000 [73.000, 150.000]
Width	ad	386	73	460.000 [126.000, 468.000]
Aratio	nonad	1983	837	1.724 [1.000, 4.345]
Aratio	ad	386	73	7.800 [2.222, 7.800]

Table 3 shows strong concentration at particular ad dimensions, consistent with repeated standardized representations in this benchmark. Ads tend to be wider and have larger aspect ratios among observed measurements. Their height has a degenerate interquartile range at 60, so an ordinary symmetric-normal description would hide important structure. Observed height means are 67.44 for ads and 63.36 for nonads; a small mean difference coexists with markedly different distributions.

Missingness is class-associated descriptively: height is unknown for 15.90% of ads versus 29.43% of nonads. Excluding every incomplete row would therefore change class composition and discard over a quarter of the data. Missingness indicators allow a predictive model to use availability information, but that information could shift when a different extractor or website population is used.

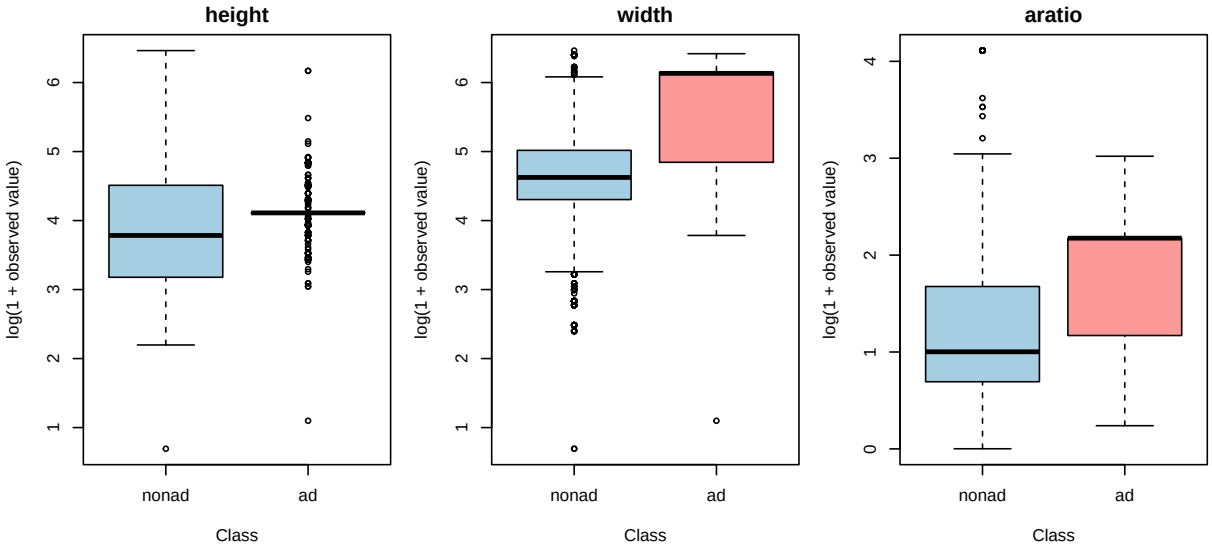


Figure 1: Class-conditional geometry on a log(1+value) scale; unknown values omitted.

Figure 1 emphasizes skewness, ties, and distributional overlap. Width and aspect ratio distinguish groups more clearly than height, but no single geometric cutoff perfectly separates the classes. The plots describe the full dataset; they were generated for reporting after the fixed model specification and are not used to select variables or tune the models.

The binary locality mean is 0.775 for observed ads and 0.766 for observed nonads, much less separated than width. The feature dictionary's 1,554 text/context indicators offer many potential distinctions, but high dimension and sparse activation make unregularized exploratory significance

scanning particularly prone to instability. No mass screening of these indicators is used to select the model.

4.1 Context-feature distributions and training support

Table 4: Mean active context indicators per record, by class, over the supplied dataset.

Family	Nonad	Ad
url	3.612	3.900
origurl	3.719	2.699
ancurl	2.655	10.340
alt	0.696	2.242
caption	0.089	0.383

Table 4 reveals that anchor-URL and alternative-text encodings are much more active among ads. These are counts of encoded indicators, not word counts or independent pieces of evidence: overlapping phrases can activate together. The summaries were computed after evaluation for interpretation and were not used to select features. They do not attribute the prediction gain to individual families; that would require a separately specified ablation analysis.

In training, 18 of 1,555 binary predictors never activate, and 241 activate at most five times. Ten test rows activate a binary feature absent from training. Such features cannot have their positive-state effect learned from that training sample. Global nonconstancy therefore does not establish useful training support. Rare terms and overlapping URL phrases also make raw coefficient magnitude a poor stand-alone importance measure.

All 2,369 complete geometry records satisfy aspect ratio approximately equal to width divided by height, within absolute difference 0.00011. Aspect ratio is consequently a derived representation, not an independently manipulable measurement. The log1p transformations do not make these columns exactly linearly dependent, but an interpretation that varies ratio while holding both dimensions fixed would be physically incoherent.

4.2 Observed training geometry separation

Objective. Quantify observed training separation without assuming independent rows for significance testing.

Table 5: Descriptive training geometry separation; no population significance claim.

Feature	Observed ad/nonad n	Ad/nonad medians	Rank separation
height	242 / 1223	60.000 / 40.000	0.640
width	242 / 1224	468.000 / 100.000	0.841
aratio	242 / 1218	7.800 / 1.649	0.708

Result. Width has the largest observed rank separation (Table 5): an observed ad–nonad pair has the ad wider in about 84.1% of comparisons, with ties counted as one half.

Interpretation. These are empirical effects, weighted by repeated rows and conditional on available measurements. They do not establish a population median difference or remove website dependence. A first-row-per-signature label reduction is not used for inference because conflicting signatures make that reduction arbitrary.

Conclusion. Geometry is associated with class in the observed training sample, especially width. Predictive value must be assessed through the withheld evaluation, not inferred from a small p-value.

5 Inferential and predictive results

5.1 Model selection: objective, result, interpretation, conclusion

Objective. Choose a ranking model without using test outcomes. All model families and grids are specified in Section 3; no further model search followed the final test evaluation.

Table 6: Training grouped cross-validation selection, using mean fold average precision.

Model	Parameter	Mean AP	Fold SD	Fold range
geometry ridge	1.0000	0.5944	0.0999	0.4660–0.7009
full ridge	0.5623	0.9196	0.0176	0.9045–0.9479
tree	0.0010	0.7913	0.1199	0.6348–0.9362

The fold SD describes variation across these particular folds; it is not a standard error from independent trials. Tree performance varies substantially more than the full ridge in this development comparison. Sixteen ridge penalties are within 0.01 mean AP of the selected full-model value, so the precise selected penalty should not be interpreted as a sharply determined scientific optimum.

Result. Figure 2 displays the recorded training and validation searches. Fold envelopes are ranges, not confidence bands. The full-feature ridge model wins the training-CV AP criterion recorded in the analysis (Table 6). The tree's chosen complexity is at the least-pruned grid boundary, so the study does not claim to have exhausted tree performance. For each family, the available grid is a finite comparison, not a global optimum search.

Interpretation. Context features appear useful for ranking beyond geometry/locality. Unequal search grids and the limited tree specification mean this is not a definitive comparison of all possible algorithms. Ridge scaling uses fitting-sample variance, including for rare binary indicators; this changes the effective shrinkage of rare features.

Conclusion. Freeze full-feature ridge as the primary ranking model. A different choice of primary utility, such as F1 at 0.5 or false-positive cost, could favor a different model. Test outcomes must not be used to retroactively redefine that criterion.

5.2 Development behavior

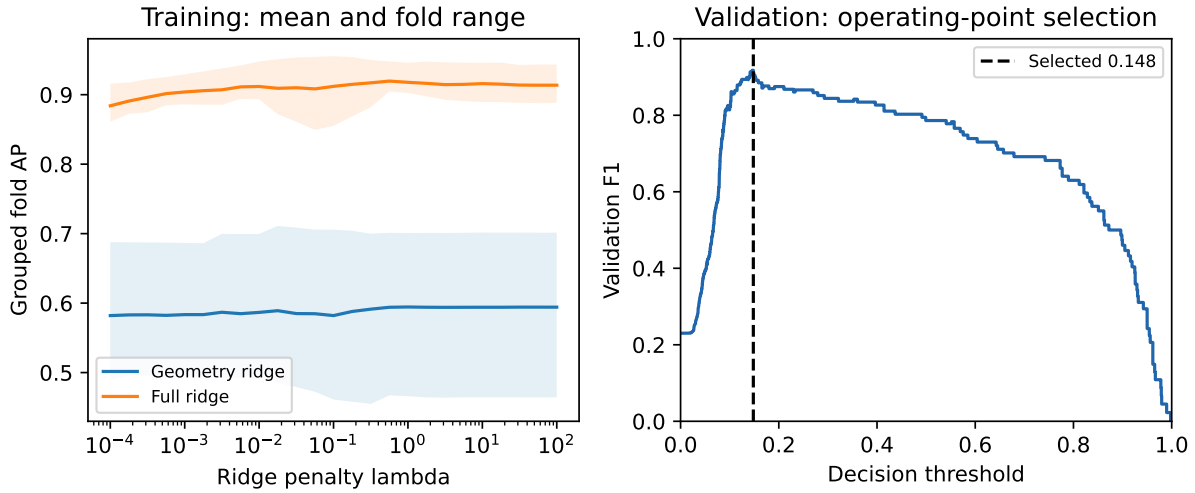


Figure 2: Recorded development searches: ridge mean fold AP with min–max fold envelopes, and validation F1 over full-ridge thresholds.

Figure 2 shows how penalty strength changes fit and how the chosen operating point sits on the validation threshold response. Shading is a fold range, not a confidence band. The left panel uses training CV and the right uses validation only; neither adds test-derived tuning. A broad near-best penalty region means the selected numerical penalty is not a sharply identified scientific optimum.

5.3 What the fitted models actually do

Let h, w, r denote the imputed log1p geometry values, l the imputed locality, and m_h, m_w, m_r, m_l the four missingness flags. On the original transformed feature scale, the compact comparator has fitted logit

$$\begin{aligned}\widehat{\eta} = &-3.02745 + 0.06738h + 0.17689w + 0.15033r - 0.00161l \\ &-0.05220m_h - 0.05213m_w - 0.05370m_r + 0.24059m_l\end{aligned}$$

This equation shows why a ranking score need not cross 0.5 even when groups differ descriptively: the intercept and regularized contributions jointly determine the probability. Coefficients refer to transformed, imputed features and are not causal effects.

The fitted tree has seven splits and eight leaves. Its first split is at log1p width 6.053181, corresponding to source width about 424.46; its wider branch splits again near source width 484.92. In the narrower branch, indicators corresponding to `ancurl*com`, `ancurl*click`, `url*ads`, `url*ad`, and `ancurl*home` distinguish leaves. A wide-image leaf contains 132 training cases with estimated ad probability .9773. The largest low-score leaf contains 1,714 cases with estimated probability .0280. The leaf counts show how much training support each rule has.

The full ridge has 1,544 nonzero fitted slopes. Ridge shrinks rather than generally selects variables, so this count is not evidence of 1,544 independent discoveries. The concrete feature-signature failure below illustrates behavior more directly than a ranking of raw coefficient magnitudes.

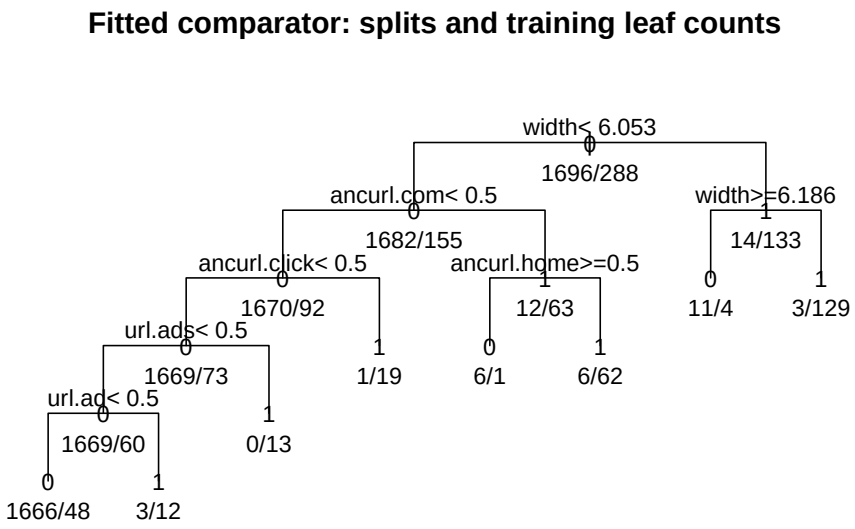


Figure 3: Actual fitted tree with transformed-width splits and training class counts at leaves.

Figure 3 displays the interaction of width and URL conditions described above. Width is on the log1p scale; leaf counts describe training support, not independent validation.

5.4 Threshold selection and final test evaluation

Objective. Quantify held-out prediction and the operating-point trade-off after selecting the model. Validation F1 is maximized at threshold 0.1481484, yielding validation F1 0.9176. This is a tuned development result, not final evidence of generalization.

Table 7: Held-out test results; ad is positive. Model selection did not use this table.

Model / operating point	Accuracy	Precision	Recall	F1	ROC AUC	AP
Always nonad	0.8658	0.0000	0.0000	0.0000	0.5000	0.1342
Geometry ridge, .5	0.8658	0.0000	0.0000	0.0000	0.7626	0.4626
Full ridge, .5	0.9457	0.9808	0.6071	0.7500	0.9569	0.8625
Tree, .5	0.9553	0.9118	0.7381	0.8158	0.8723	0.7217
Full ridge, tuned	0.9329	0.7283	0.7976	0.7614	0.9569	0.8625

Result. The final tuned ridge correctly identifies 67 ads and 517 nonads, missing 17 ads and falsely flagging 25 nonads. Balanced accuracy is 0.8757; specificity is 0.9539. The raw test accuracy is lower than that of the tree, and the tree also has higher test F1 at its fixed 0.5 threshold (Table 7). Full ridge nevertheless has better ranking AP and ROC AUC in this comparison.

Interpretation. There is no uniformly best model across metrics. Geometry ridge ranks ads above nonads moderately well but makes no ad predictions at 0.5: discrimination and a fixed decision threshold are different questions. Lowering full-ridge's threshold raises recall from 0.607 to 0.798 while reducing precision from 0.981 to 0.728. The gain in recall therefore needs to be considered alongside the additional false alarms.

Conclusion. Full-context information materially improves held-out ranking over the geometry/locality comparator. Lowering the ridge threshold detects more ads, but the additional false positives largely offset that gain in F1. Validation F1 of 0.918 falls to 0.761 on test; the error analysis below examines the sample patterns behind this difference.

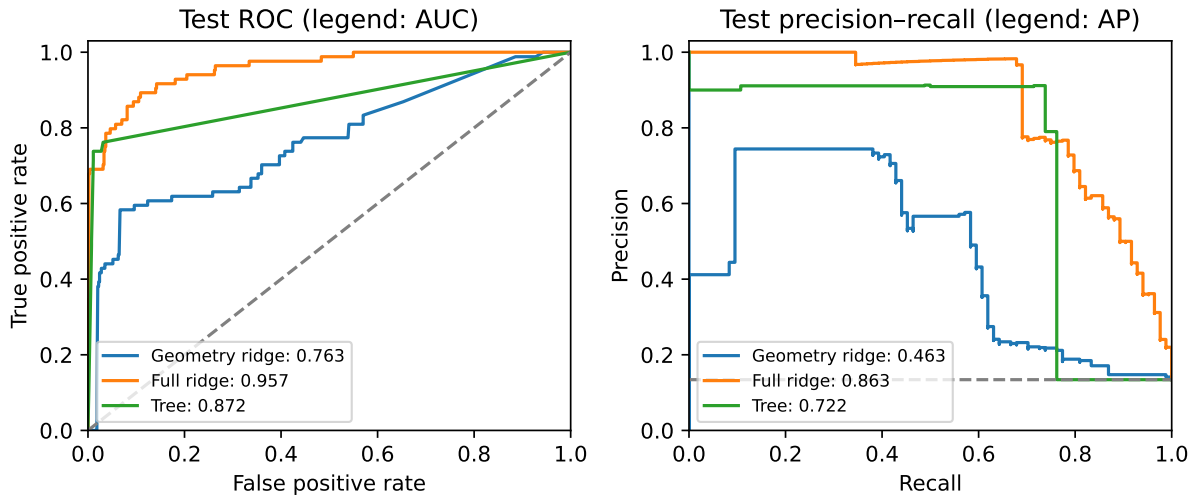


Figure 4: Held-out ranking curves for all three fitted comparators. Legends report AUC and AP; dashed lines show random-ranking ROC and test-prevalence PR baselines.

Figure 4 connects the ranking and classification results. The tree assigns only eight distinct test scores, compared with 484 for full ridge. Of the 45,528 possible test ad–nonad pairs, 10,645 receive tied tree scores. Its leaves therefore make useful decisions at .5 while grouping many images at the same probability, limiting ranking resolution. Full ridge separates scores more finely and achieves higher AP, yet its validation-selected threshold admits more nonads. Geometry ridge illustrates the opposite threshold problem: it ranks cases usefully but all scores remain below .5. Ranking, probability estimation and thresholded classification are consequently three different aspects of performance. Tied scores enter together; AP .8625 is the sum over recall increments, not the trapezoidal area of a smoothed line. The PR display exposes false alarms among retrieved candidates more directly than ROC when nonads are common.

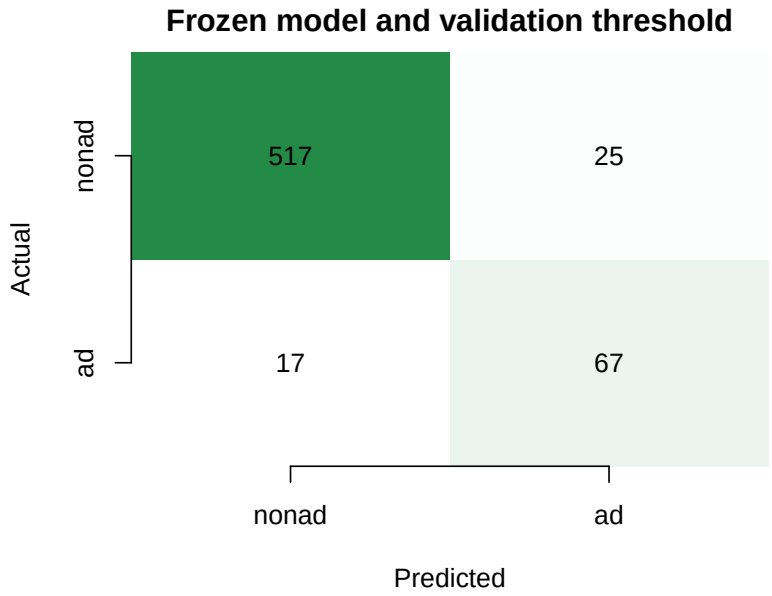


Figure 5: Test confusion counts at the validation-selected threshold 0.1481484.

Figure 5 makes the remaining errors explicit: 25 false alarms and 17 misses. The 84-positive test sample is small enough that a few additional errors would materially change precision and recall.

5.5 Post-evaluation failure analysis

This subsection was developed after inspecting test results. It explains the fixed predictions; it does not select a replacement model, remove difficult cases, or change the threshold.

The 42 test errors occupy 27 distinct predictor-pattern groups. One nonad signature occurs 15 times and receives score .360503 on every occurrence, accounting for 60% of the 25 false positives and 35.7% of all errors. Its three geometric measurements are unknown, locality is one, and its active terms include repeated `keith`, `dumble`, `members` and `ukonline.co.uk` signatures across URL families. These observations describe the representation; without pixels or original pages they cannot establish why the image was labeled nonad or attribute the prediction to one phrase.

For supplied row 844 (exported index 843), missing log1p geometry is imputed to 3.931826, 4.663439, and 1.127167. Locality and geometry-missingness flags are one. The geometry ridge gives logit −1.927774 and probability 0.126997. The full ridge gives logit −0.573180 and probability 0.360503. The full-model probability exceeds the frozen threshold 0.1481484, so the decision is ad despite the supplied nonad label. The score difference between separately fitted models cannot be attributed to one context phrase.

Keeping this group is essential to the row-weighted evaluation target. Its repeated contribution also makes precision sensitive to cluster composition. Excluding it would change the target and make the performance look better; it is not a correction.

Table 8: Frozen-threshold performance by geometry availability; post-hoc descriptive slices.

Partition	Geometry	Rows	Ads	TP	FP	F1
validation	Observed	472	73	68	5	0.932
validation	Missing	197	14	10	0	0.833
test	Observed	437	71	61	8	0.871
test	Missing	189	13	6	17	0.333

Table 8 identifies a much weaker test operating point when geometry is missing. The subset has only 13 ads, and its false positives overlap the repeated signature, so these are not independent explanations or a causal effect of missingness. Validation has a similar number of incomplete cases and ads but not the same signatures. Availability counts alone therefore do not explain transfer of threshold performance.

The validation decisions are unchanged throughout approximately $(.1456895, .1481484]$. Thus .1481484 identifies a selected observed score, not a uniquely meaningful probability boundary. Validation F1 is .818 at threshold .1, .918 at the selected threshold, .875 at .2 and .786 at .5. The chosen point exploits a favorable validation ordering that need not persist in another group sample. This supports reporting a ranking score as the primary outcome and treating the binary decision as an illustrative, validation-selected policy.

5.6 Conditional uncertainty and probability assessment

Objective. Estimate test-sampling uncertainty without treating repeated predictor rows as independent, and inspect whether high ranking implies good probability estimates.

Table 9: Cluster-bootstrap percentile intervals for the frozen model and threshold (1,000 resamples).

Quantity	Point estimate	Conditional 95% interval
accuracy	0.9329	[0.8799, 0.9675]
balanced accuracy	0.8757	[0.8181, 0.9255]
precision	0.7283	[0.5098, 0.9296]
recall	0.7976	[0.7011, 0.8847]
F1	0.7614	[0.6176, 0.8791]
ROC AUC	0.9569	[0.9277, 0.9807]
AP	0.8625	[0.7840, 0.9320]
Brier	0.0467	[0.0341, 0.0601]
AP gain	0.4000	[0.2053, 0.5418]

Result. Full ridge improves AP over geometry ridge by 0.4000, with a conditional 95% interval of [0.2053, 0.5418] (Table 9). This is a gain in AP units, not a 40% increase in accuracy. The entire interval is positive, supporting a substantial ranking improvement under the cluster-sampling model. Precision has a much wider interval, [0.5098, 0.9296], so the proportion of correct ad predictions at the selected threshold is less precisely estimated.

The pairing uses the same resampled groups for both models, comparing their performance on the same easy and difficult cases. A 95% confidence procedure is intended to cover the population quantity in about 95% of repeated samples under its assumptions. Interval width expresses estimation precision; it is not the range containing 95% of individual predictions.

Interpretation. These intervals describe a fixed predictor and threshold, conditional on the development data and an exchangeable-pattern-group approximation. They are not confidence statements for retraining on arbitrary future samples, and do not account for site-level dependencies. No strong distributional approximation for F1 is imposed, but bootstrap inference with only 84 ads and an unknown sampling frame remains approximate.

Conclusion. The measured full-versus-geometry ranking gain is supported within this split; the practical precision of the tuned classifier remains uncertain. Retaining one representative per test pattern gives AP 0.8843, ROC AUC 0.9584 and F1 0.8188 across 484 patterns. This unplanned sensitivity summary is not a replacement primary test, but shows that row-weighted

and pattern-weighted targets differ.

The Brier score is 0.0467 for ridge, 0.0435 for the tree and 0.1163 for the training-prevalence constant score. Thus the selected ranking model does not have the best Brier score among these comparators. Coarse test bins suggest probability miscalibration but do not isolate it from discrimination or sampling variation: scores above 0.4 correspond to 57 ads among 58 observations, while the lowest bin contains 18 ads among 540 observations. These are descriptive, coarse-bin observations, not a calibration proof. Recalibrating on the test outcomes would invalidate their held-out role and is not done.

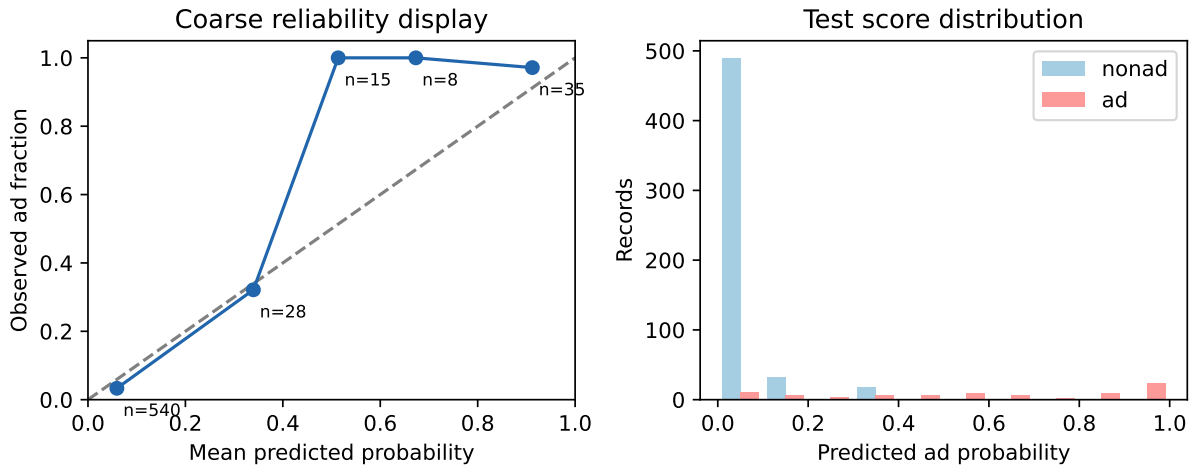


Figure 6: Post-evaluation probability assessment: reliability bins with record counts and the class-specific score distribution.

Figure 6 makes the bin sizes visible. The high observed fractions above score .4 suggest underprediction in those bins, but there are only 15, 8 and 35 records in the three upper bins. The lowest bin contains 540 records. Connecting bin averages is a descriptive display, not an estimated calibration function or an independent-sampling confidence statement. The histogram distinguishes concentrated low nonad scores from sparse high-score regions. No test-based recalibration is applied.

6 Discussion, limitations, and extensions

6.1 Answers to the research questions

1. What distinguishes the geometric measurements? Width provides the strongest observed separation: in training, its rank separation is .841, versus .640 for height. Ads concentrate at particular dimensions, whereas nonads have a wider variety of shapes. Height means alone obscure this structure: the ad height interquartile range is concentrated at 60 even though its mean is close to the nonad mean. Missing geometry is also more common among nonads, so complete-case summaries describe a different class mix from the full sample.

2. What do context features add? Test AP rises from .4626 for geometry/locality ridge to .8625 for full ridge. The paired interval for the .4000 gain is entirely positive. Thus the contextual representation supplies substantial information beyond shape and locality. Greater anchor-URL and alternative-text activity among ads helps describe that representation, although the comparison evaluates the whole added feature set rather than individual families.

3. Which model performs best? Full ridge leads on AP and ROC AUC, while the tree has higher observed F1 (.8158 versus .7614 for tuned ridge) and lower Brier score. The tree's discrete leaves and ridge's finer scores explain why these rankings need not agree. Geometry ridge's zero recall at .5 shows why judging a model only by its default threshold would miss its ranking ability. Model choice therefore follows the analytical objective: this study selected ranking by AP, then examined a separate classification threshold.

4. What does uncertainty reveal? The positive AP-gain interval supports the central comparison, but precision and F1 intervals are wide. Error composition explains why: one repeated nonad signature contributes 15 of the 25 false positives, and missing-geometry test cases have much poorer F1. These observations show how a relatively small number of patterns can affect the apparent success of a threshold.

6.2 Strengths and limitations

The study combines an interpretable geometry comparator, a regularized full-feature model and a nonlinear tree. Keeping identical signatures together avoids evaluating a model on copies of training representations; a separate validation set chooses the threshold before test evaluation. The fitted equation, tree, worked prediction and generated result tables make the analysis reproducible and connect numerical results to actual model behavior.

Three limitations matter to interpretation. First, the supplied matrix lacks original images, page identifiers and detailed annotation records. Different signatures may share a website, and conflicting labels may reflect annotation differences or information absent from the features. Second, the intervals assume independent signature clusters and hold fitted models and tuning choices fixed; they measure test-sample uncertainty, not variation from repeating the entire study. Third, sparse correlated indicators and class-associated missing geometry complicate interpretation: fitted slopes are associations, and the larger model comparison does not isolate individual feature families. These limitations qualify the statistical interpretation without changing the observed comparisons.

6.3 What would justify an action?

On the fixed test sample, changing full ridge from .5 to the selected threshold recovers 16 additional ads but creates 24 additional false positives. Let c_{FN} be the loss from a missed ad and c_{FP} the loss from falsely blocking a nonad. The observed loss change is

$$L_{tuned} - L_{.5} = 24c_{FP} - 16c_{FN}.$$

The tuned policy has lower observed loss than .5 only when $c_{FP}/c_{FN} < 2/3$. This retrospectively describes two already-reported policies; it is not a test-selected rule or population break-even estimate. It shows why F1 optimization can conflict with avoiding false blocking.

If p were calibrated for deployment, predicting ad has conditional loss $c_{FP}(1-p)$ and predicting nonad has loss $c_{FN}p$. Minimum loss requires $p > c_{FP}/(c_{FP}+c_{FN})$. This explains why a threshold chosen by F1 need not minimize a particular cost: the two objectives weight mistakes differently.

6.4 Targeted extensions

Two extensions follow directly from the findings. A development-data comparison that adds one context family at a time could clarify which families contribute most to ranking. A new grouped evaluation could then examine whether the missing-geometry weakness and repeated-signature

errors recur. Any revised feature set or threshold should be evaluated on new held-out data rather than selected to improve the test results already examined here.

6.5 Final conclusion

Web-context features substantially improve advertisement ranking over geometry and locality alone: full ridge achieves test AP .8625 versus .4626, with a paired 95% interval of [.2053, .5418] for the gain. Width is the most informative geometric measurement, but geometry alone does not produce useful ad decisions at the default .5 threshold. The tree achieves better test F1 at its reported operating point, demonstrating that stronger ranking does not automatically mean stronger thresholded classification. Finally, the concentration of false positives in one repeated signature explains why overall metrics should be read alongside the actual errors. Together, these results answer the project's questions about feature information, model behavior and estimation uncertainty.

7 Data, code, and reproducibility sources

- Original dataset and attribution: https://archive.ics.uci.edu/dataset/51/internet+advertisements; DOI https://doi.org/10.24432/C5V011. Repository license: CC BY 4.0.
- Original downloadable files: https://archive.ics.uci.edu/static/public/51/internet+advertisements.zip. Copies of `ad.names`, `ad.DOCUMENTATION`, and `ad.data` are retained under `analysis/source/`.
- Supplied analysis input: `data/add.csv`; matched numerically and by label to the original source file.
- Executed R code: `analysis/run.R`. Independent check: `analysis/verify.py`. Public R source: https://github.com/hlbathien/internet-advertisements-study/blob/main/analysis/run.R. The repository contains the reproducible code and report sources.
- Execution instructions: `README.md`. Split/fold assignments, test predictions, metric tables, model objects, and session record are under `analysis/results/`.

Run from the project root: `Rscript analysis/run.R`, then `python3 analysis/verify.py`. Regenerate the post-evaluation summaries and figures without fitting models:

```
python3 analysis/critique_diagnostics.py
python3 analysis/plot_calibration.py
Rscript analysis/export_models.R
python3 analysis/render_tables.py
```

R dependencies are `glmnet` and `rpart`; verification and plots use Python with NumPy, pandas, scikit-learn and Matplotlib. The executed R session used R 4.1.2, glmnet 4.1-3, rpart 4.1.16 and Matrix 1.4-0. Exact session details are saved in `analysis/results/sessionInfo.txt`. The PDF is built by running pdfLaTeX twice in `report/`. Absolute machine-specific input paths are not used by the analysis.

8 References

1. Friedman, J., Hastie, T., and Tibshirani, R. *An Introduction to glmnet*. Official glmnet documentation. https://glmnet.stanford.edu/articles/glmnet.html.
2. Kushmerick, N. (1999). *Internet Advertisements* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5V011. Original accompanying files: `ad.DOCUMENTATION`, `ad.names`.
3. Kushmerick, N. (1999). *Learning to remove Internet advertisements*. Proceedings of the Third Annual Conference on Autonomous Agents, 175–181. https://doi.org/10.1145/301136.301186.
4. R Core / rpart documentation. *Control for Rpart Fits*. https://stat.ethz.ch/R-manual/R-devel/library/rpart/html/rpart.control.html.
5. Scikit-learn developers. *average_precision_score*. Official API documentation. https://scikit-learn.org/stable/modules/generated/sklearn.metrics.average_precision_score.html.